



Lower and Higher Order Thinking Skills (LOTS and HOTS)

Blooms Taxonomy

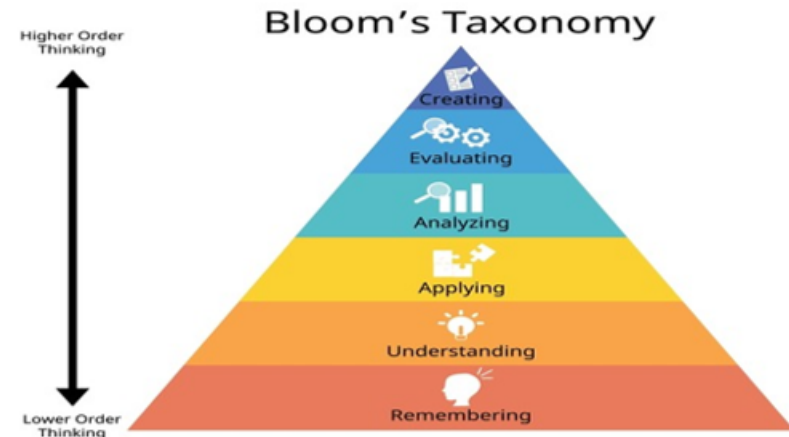
Dr. Mayada Mahmoud

DEFINITION OF Bloom's Taxonomy

It is a comprehensive framework used to classify and organize educational goals, objectives, and outcomes.

- ✓ It was initially developed by educational psychologist Benjamin Bloom in 1956.
- ✓ The taxonomy is ranging from simple recall of facts to more complex processes like critical thinking and creativity.
- ✓ The taxonomy is a hierarchical pyramid, with each level building on the previous one.

Dr. Mayada Mahmoud

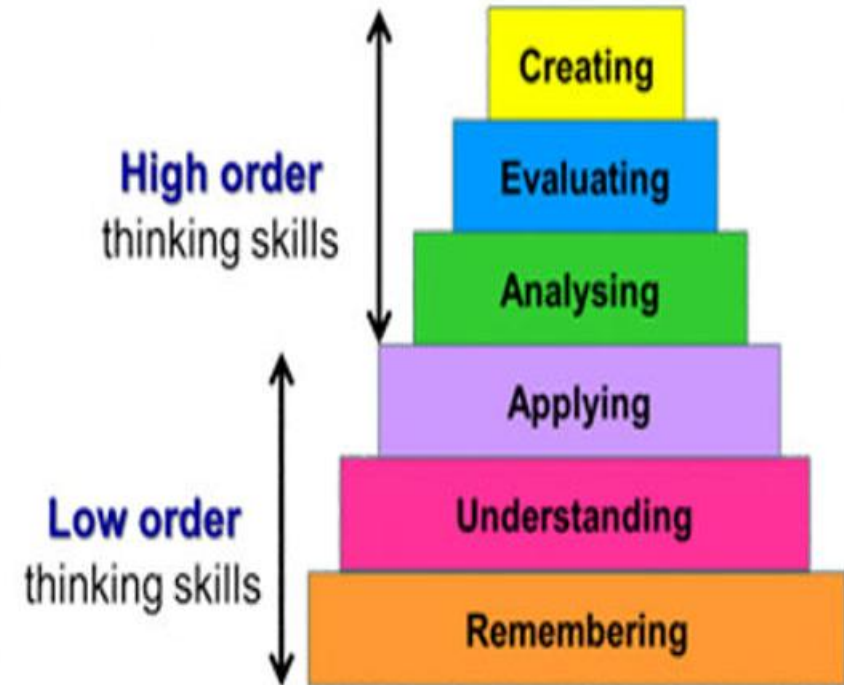


Importance OF Bloom's Taxonomy

Fostering Critical Thinking

- Support [higher-order thinking skills](#):
1. **Analyze**: Break down complex information into parts and examine relationships.
 2. **Evaluate**: Make informed judgments based on criteria and standards.
 3. **Create**: Synthesize information to produce original ideas and solutions.

These skills for problem-solving and decision-making.



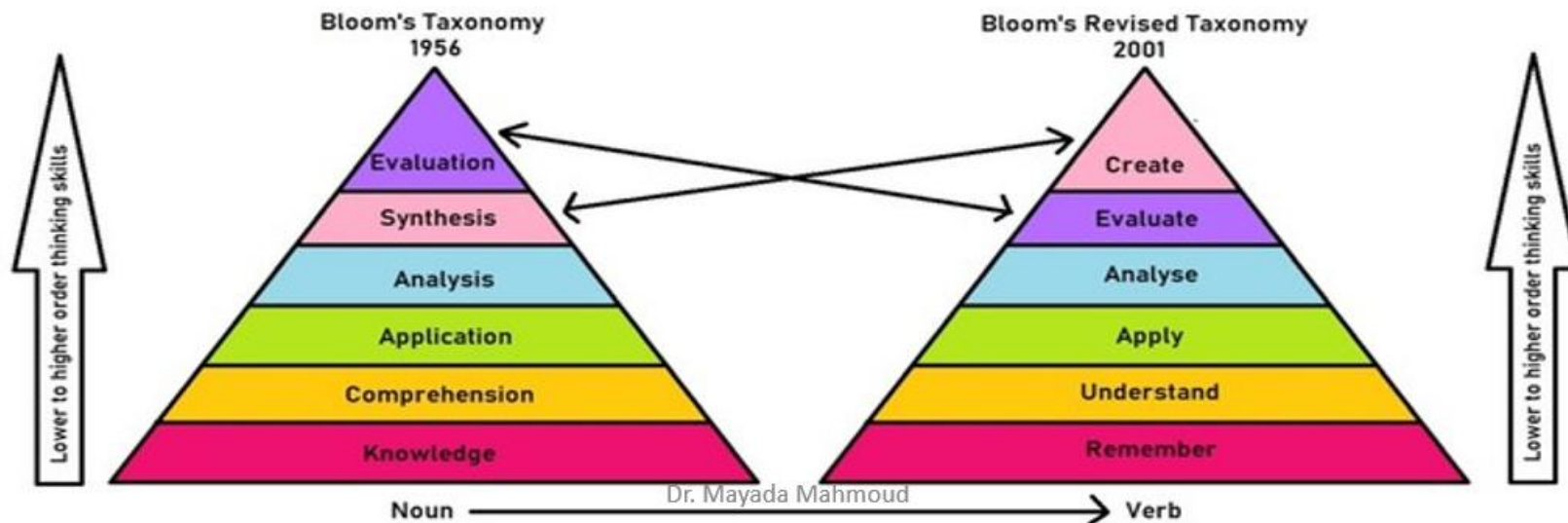
Models of Bloom's Taxonomy

1. The original taxonomy comprised six levels:

- Knowledge, Comprehension, Application, Analysis, Synthesis, and Evaluation.

2. Revised version comprised six levels:

- Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating.



The Original **Bloom's Taxonomy** (1956)

- Categorizes cognitive skills into **six levels**:

1. **Knowledge**: Recalling facts and basic concepts.
2. **Comprehension**: Understanding information and grasping meaning.
3. **Application**: Using knowledge in new situations.
4. **Analysis**: Breaking down information into parts to explore understandings and relationships.
5. **Synthesis**: collecting information in a different way to propose alternative solutions.
6. **Evaluation**: Justifying a decision or course of action through criteria and standards.



- This progression from knowledge to evaluation reflects a move from simple recall of information to higher-level thinking, such as critical analysis and judgment.

Revised Bloom's Taxonomy

A Modern Approach(2001)



- 1. Remembering:** Retrieving relevant knowledge from long-term memory. This level emphasizes the importance of recalling facts and basic.
- 2. Understanding:** Constructing meaning from messages, interpreting, classifying, summarizing, inferring, comparing, and explaining information.
- 3. Applying:** Using a procedure in a given situation or the practical utility of knowledge.
- 4. Analyzing:** Breaking material into small parts and detecting how the parts relate to one another. This level used to identify motives, causes, and principles.
- 5. Evaluating:** Making judgments based on criteria and standards. It is fostering a capacity for critical assessment and decision-making.
- 6. Creating:** Putting elements together to form a novel, to make an original product. This skill generate new ideas which encourages innovation and creativity.

Older version **VS** Revised Version

Older version

Revised Bloom's taxonomy

Knowledge

Remember

Comprehension

Understand

Application

Apply

Analysis

Analyze

Synthesis

Evaluate

Evaluation

Create

Gap Filling Activities (remembering skill)

- Fill in the blanks with the words from the list. You may need to change the forms of the words. You may need to use one word more than once.
- Writing letters is a part of our social activities. It is (a) — form of communication, which plays a vital role in our practical life. (b) — this time. (c) — revolution has taken place in the field of (d) — this like telephone, wireless, cellular phone, fax and most recently (e) —, which are much faster means (f) — communication. Letter writing is (g) — kind of good art which is (h) — important means (i) — maintaining good relations (j) — people, offices and business organizations.
- **Clues:** *the — for — on — communication — an — of — e-mail — by — a — with*

Answer

- a) a
- b) By
- c) a
- d) communication
- e) e-mail
- f) of
- g) a
- h) an
- i) of
- j) with.

Match the following

- A) Circulatory System**
- B) Digestive System**
- C) Endocrine System**
- D) Immune System**
- E) Integumentary System**
- F) Lymphatic System**
- G) Muscular System**
- H) Nervous System**
- I) Reproductive System**
- J) Respiratory System**
- K) Skeletal System**
- L) Urinary System**

- 1) The System that regulates the body's responses to internal and external stimuli
- 2) The System that differentiates self from non-self and neutralizes potentially pathogenic organisms or substances
- 3) The System consisting of the skin and its associated structures, such as the hair, nails, sweat glands, and sebaceous glands
- 4) The System involved in reproduction
- 5) The System of organs that produce, collect, and eliminate urine
- 6) The framework of the body, consisting of bones and other connective tissues, which protects and supports the body tissues and internal organs
- 7) The System of endocrine glands that chemically controls the various functions of cells, tissues, and organs through the secretion of hormones
- 8) The System of organs and glands responsible for the ingestion, digestion, and absorption of food
- 9) The System that circulates blood and lymph through the body
- 10) The System of organs involved in the intake and exchange of oxygen and carbon dioxide between the body and the environment
- 11) The System that is composed of skeletal, smooth, and cardiac muscle tissue and functions in movement of the body or of materials through the body, maintenance of posture, and heat production
- 12) Part of the immune System; the System which circulates lymph